

ASSIGNMENT #3

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
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Due Date: Wednesday January 22 by 5:00pm in my office

Number of points.....100

Your score.....~~84~~ 63

Note: All problems are worth 100 points. Your final score will be the average value

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Introduction

The objective was to calculate the process gains for optimal controller tuning with a 3-inch equal percentage valve. To control the temperature of the reactor, the following control methods were performed: feedback, feedback combined with cascade control and feedback combined with feed forward control. For the purpose of this assignment, the following assumptions were made:

1. Constant density
2. Constant heat capacity
3. Constant pressure
4. Assume perfect mixing in the CSTR
5. Uniform level

All tests were down by step up, step up then step back to steady state value.

Data and Equations

Table 1: Data and Results for Acetic Acid with Ethanol CSTR

Parameters	Feed	Product
Density, kg/m^3	930.37	870.631
Heat Capacity $J/kg \cdot K$	1.913	2.511
Heat of Reaction, kJ/mol	-29.069	
UA, $J/min \cdot K$	575634.67	
ΔP , psi	7.26001	

Density and Heat Capacity were retrieved from the properties in Aspen.

Heat of Reaction:

Heat of reaction = enthalpy of pure components(products – reactants)

$$(-282.38 + -471.834) \text{ kJ/mol} - (-453.51 + -271.63) = -29.069 \text{ kJ/mol} \quad [1]$$

Overall Heat Transfer Coefficient:

$$Q_{cool} = UA(T_R - T_c) \quad [2]$$

$$UA = 431726 \frac{J}{sec} * \frac{60 \text{ sec}}{1 \text{ min}} * \frac{1}{45 \text{ K}}$$

Pressure Change:

$$Q_{cool} = C_v \sqrt{\frac{\Delta P}{sg}} \quad [3]$$

$$97 \text{ gpm} = 36 * \sqrt{\frac{\Delta P}{1}}$$

Set Points for All Controllers

For the control scheme given, there ^{are} 3 set points. One setpoint being the reactor temperature, the other being the cooling water outlet temperature and the other being the percent of the valve opening. The percent of the valve opening is at steady state of 60%. The reactor temperature setpoint should be set to 70°C and the cooling water outlet temperature should be set to 42°C. The reactor temperature setpoint was based off the basis of Homework 1. When setting the reactor to 70°C and the inlet of the cooling water at the given parameters, the outlet for the cooling water temperature was given at 42°C. That value then became the setpoint value for the cooling water outlet.

Have to specify which process these belong to, missing or also.
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Process Gain for Feedback Controller

The step up, step down and step back test consisted of changing the opening percentage from steady state (60%), then stepping up to 70%, stepping down to 50% and then stepping back to 60%. The flowrates were calculated at each of the percentages by using equation 3.

$$K_{P,FB} [=] \frac{\Delta^{\circ}\text{C}(\text{Reactor Temperature}, T_R)}{\% \Delta(CO)}$$

Table 2: Parameters for Feedback Controller

% Opening	Q (gpm)	T_R (°C)	%Δ Opening	ΔT_R (°C)
70	162.74	54	10	16
50	58.47	76	20	22
60	97	70	10	6

The first data point in the ΔT_R column is the difference between the steady state value and the step up. The second point, between the step up and step down. The third point, between the step down and step back.

Table 3: Process Gain for Feedback Controller

$K_{P,FB}$	Average $K_{P,FB}$
1.6	1.1
1.1	
0.6	

Coding for Feedback Controller

!Variable Declarations and Data

Parameter Rg=8.3145 As Real !Universal Gas Constant, J/(mol K)

Global Q, CAAfeed, CETOHfeed, CEACfeed, CWATERfeed, Tfeed As Real

Q = 454.249 !L/min

CAAfeed = 8.81209 !mol/L

CETOHfeed = 8.52726 !mol/L

CEACfeed = 0 !mol/L

CWATERfeed = 0.462857 !mol/L

Tfeed = 298 !K

Global Cv, deltaP, K, Qcool As Real

Cv = 36

deltaP = 7.26001 !psi

K = 3.78541 !gal/L

Qcool = K*Cv*sqrt(deltaP) !L/min

Global Vol, UA, Tc, UAWater, VolWater As Real

Vol = 10000 !L

UA = 575634.67 !J/(min K)

Tc = 298 !K

UAWater = 1514167.5 !J/(min K)

VolWater = 13.48 !L

Global Rho, Cp, DHR1, CpWater, RhoWater As Real

Rho = 930.37 !g/L

Cp = 2.511 !J/(g K)

DHR1 = -29069.0 !J/mol

CpWater = 74.2168 !J/(mol K)

RhoWater = 55.173 !mol/L

Global kfo, kro, ERfo, ERro As Real

kfo = 1.14e10 !mol/(L*min)

kro = 3e9 !mol/(L*min)

ERfo = 7156.17 !K

ERro = 7156.17 !K

@Initial Conditions

U(1)=0 !12.4921

U(2)=0

U(3)=0

U(4)=0 !12.4921

U(5)=298

U(6)=298

@Model Equations

Dim CAA, CETOH, CEAC, CWater, Temp, TempWater As Real

CAA=U(1)

CETOH=U(2)

CEAC=U(3)

CWater=U(4)

Temp=U(5)

TempWater=U(6)

Dim kf, kr, R1 As Real

kf=kfo*exp(-ERfo/Temp)

kr=kro*exp(-ERro/Temp)

R1=kf*CAA*CETOH-kr*CEAC*CWater

If(iRange=1)Qcool = 616.038 !L/min

If(iRange=2)Qcool = 221.333 !L/min

If(iRange=3)Qcool = 367.185 !L/min

F(1)=Q*CAAfeed-Q*CAA-R1*Vol

F(2)=Q*CETOHfeed-Q*CETOH-R1*Vol

F(3)=Q*CEACfeed-Q*CEAC+R1*Vol

F(4)=Q*CWaterfeed-Q*CWater+R1*Vol

F(5)=Q*Rho*Cp*(Tfeed-Temp)+(-DHR1)*R1*Vol-UA*(Temp-Tc)*Qcool/ 367.185

F(6)=Qcool*RhoWater*CpWater*(Tc-TempWater)+UA*(Temp-Tc)*Qcool /367.185

@Coefficient Matrix

E(1)=Vol

E(2)=Vol

E(3)=Vol

E(4)=Vol

E(5)=Rho*Cp*Vol

E(6)=RhoWater*CpWater*VolWater

@Solver Options

Neq=6

Npts=50

Tend=800

Tstop=200;400;600

Param(28)=1

VariableNames=Time (min);CAA;CETOH;CEAC;CWater;Temperature (K);TempWater (K)

Graph for Feedback Controller

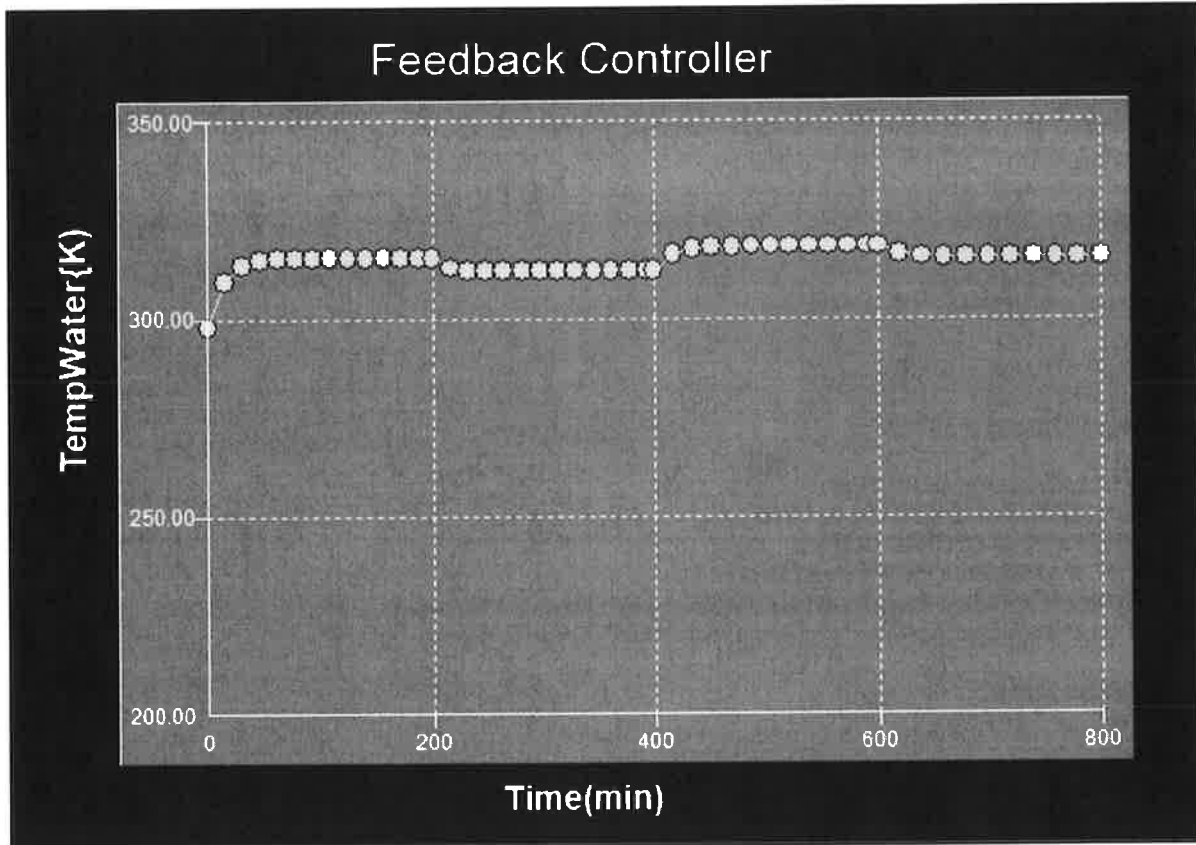


Figure 1. Stepped Process for Feedback Controller

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Process Gains for Cascade Controller

$$K_{P1,CS} [=] \frac{\Delta^{\circ}\text{C}(\text{Cooling Jacket Temperature}, T_{CJ})}{\% \Delta(CO)}$$

$$K_{P2,CS} [=] \frac{\Delta^{\circ}\text{C}(\text{Reactor Temperature}, T_R)}{\Delta^{\circ}\text{C}(\text{Cascade Set Point Temperature}, T_{SP})}$$

Plot? - 8
Cool? - 5

Table 4. Parameters for Cascade Controller, part 1

% Opening	Q (gpm)	T_{CJ} (°C)	%Δ Opening	ΔT_{CJ} (°C)
70	162.74	35	10	7
50	58.47	53	20	18
60	97	42	10	11

Table 5. Process Gain for Cascade Controller, part 1

$K_{P1,CS}$	Average $K_{P1,CS}$
0.7	0.9 - 3
0.9	
1.1	

For the second part of the cascade controller, the cascade set point temperature is the master controller. The master controller is defined as the outlet temperature of the cooling water.

Table 6. Parameters for Cascade Controller, part 2

T_{SP} (°C)	T_R (°C)	ΔT_{SP}	ΔT_R
45	66	3	4
40	73	5	7
42	70	2	3

Table 7. Process Gain for Cascade Controller, part 2

$K_{P2,CS}$	Average $K_{P2,CS}$
1.333	1.411 - 3
1.4	
1.5	

Process Gain for Feed Forward Controller

$$K_{P,FF} [=] \frac{\Delta^{\circ}\text{C}(\text{Reactor Temperature}, T_R)}{\Delta^{\circ}\text{C}(\text{Cooling Water Temperature}, T_c)}$$

Table 8. Parameters for Feed Forward Controller

T_c (°C)	T_R (°C)	ΔT_c	ΔT_R
30	76	5	6
20	63	10	13
25	70	5	7

Table 9. Process Gain for Feed Forward Controller

$K_{P,FF}$	Average $K_{P,FF}$
1.2	1.3 / 3
1.3	
1.4	

Coding for Feed Forward Controller

!Variable Declarations and Data

!*****

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@Model Equations

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CAA=U(1)

CETOH=U(2)

CEAC=U(3)

CWater=U(4)

Temp=U(5)

TempWater = 315

Dim kf, kr, R1 As Real

kf=kfo*exp(-ERfo/Temp)

kr=kro*exp(-ERro/Temp)

R1=kf*CAA*CETOH-kr*CEAC*CWater

If(iRange=1)Tc=303

If(iRange=2)Tc=293

If(iRange=3)Tc=298

F(1)=Q*CAAfeed-Q*CAA-R1*Vol

F(2)=Q*CETOHfeed-Q*CETOH-R1*Vol

F(3)=Q*CEACfeed-Q*CEAC+R1*Vol

F(4)=Q*CWATERfeed-Q*CWater+R1*Vol

F(5)=Q*Rho*Cp*(Tfeed-Temp)+(-DHR1)*R1*Vol-UA*(Temp-Tc)*Qcool/ 367.185

F(6)=Qcool*RhoWater*CpWater*(Tc-TempWater)+UA*(Temp-Tc)*Qcool /367.185

@Coefficient Matrix

E(1)=Vol

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@Solver Options

Neq=6

Npts=50

Tend=800

Tstop=200;400;600

Param(28)=1

VariableNames=Time(min);CAA;CETOH;CEAC;CWater;Temperature(K);TempWater {K}

Graph for Feed Forward Controller

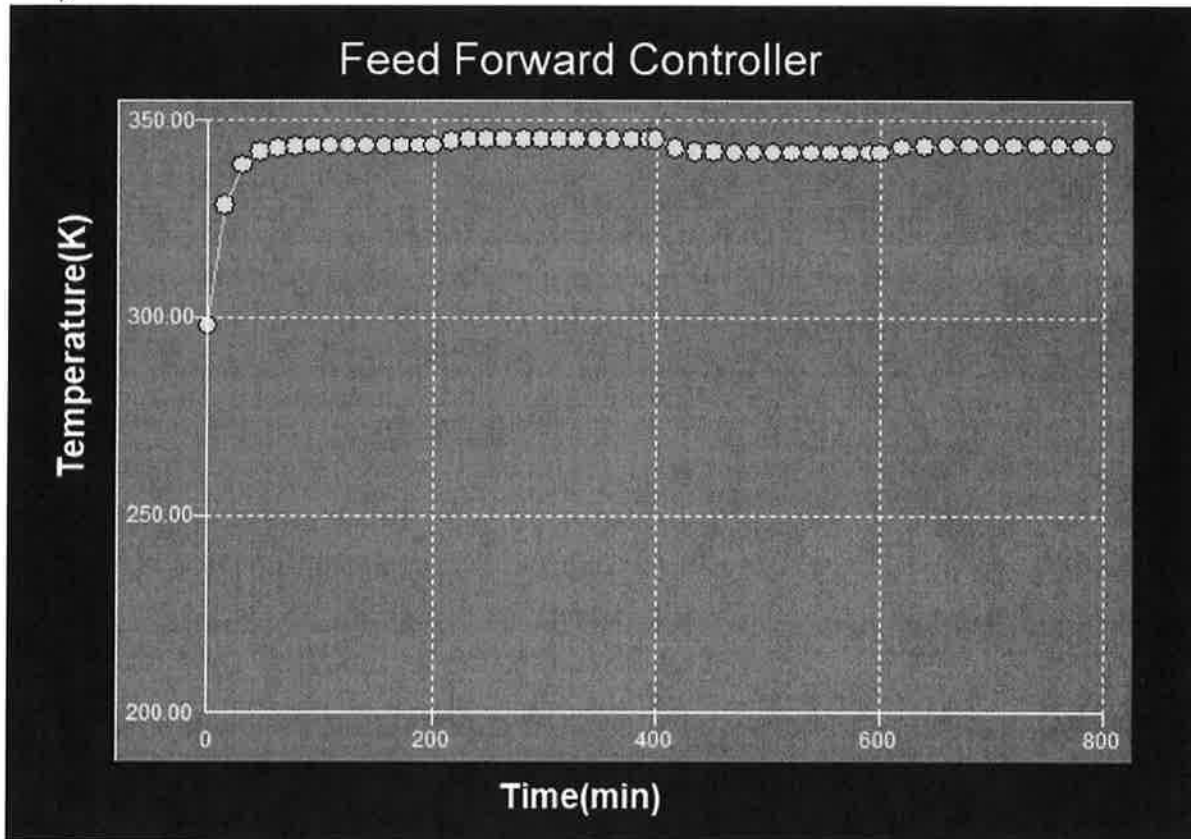


Figure X. Stepped Process for Feedforward Controller

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